MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY

INFORMATION ON RESEARCH RESULTS

1. General information:

Project code: THS2026-68

Project title: An Edge AI–Based System for Multimodal Analysis and Clustering of Flood Rescue Events

Coordinator: Lê Thị Ngọc Ảnh

Implementing institution: Trường Công nghệ Thông tin và Truyền thông, Đại học Cần Thơ

Duration: from March 2026 to August 2026.

2. Objective(s):

Develop a prototype for analyzing and consolidating flood-rescue reports under intermittent connectivity, combining edge inference, compact transmission and event clustering.

3. Creativeness and innovativeness:

Combine adaptive sending, persistent queues and event clustering with conditional edge-localization and exact-copy invariance audits. Evaluate clustering, ranking and dispatch separately.

4. Research results:

The seed-1024 image model achieves 75.78% accuracy and 74.26% macro-F1 on 256 test images, with three critical errors. A separate split manifest and runtime/compression/mobile benchmarks for this model remain unavailable. Historical benchmarks are reported separately; PTQ/QAT summary–CSV discrepancies remain unresolved. Structured urgency uses a synthetic 32-state table rather than a Vietnamese text corpus. RQ1–RQ3 do not establish general superiority or field rescue benefits.

5. Products:

Flutter application, FastAPI/SQLite backend, map dashboard, model assets, experiment data, research report and bilingual bulletins/summaries. An ISDS 2026 draft exists without publication or formal adoption evidence.

6. Effects, technology transfer means and applicability:

The prototype supports education, research and workflow demonstration. Planned transfer uses the application, source code, dashboard and operating documentation; no completed transfer record is available. Text modeling, urgency-field integration, repeated Android/network/SMS tests and expert validation remain necessary before operational deployment.